

Innovation in Chemical Engineering

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The concept of innovation goes beyond the pursuit of new products — all stakeholders need to be involved.

It seems just about everyone today is talking about innovation and how important it is for companies to grow and survive. Hundreds of books and thousands of papers have been written about innovation, but many tend to focus on techniques to generate ideas for innovative products. Most miss the basic fact that innovation goes well beyond the invention of new products.

The chemical process industries (CPI) and the chemical engineering profession are not always viewed by the general public as being very innovative fields. This perception may have some validity if we compare the CPI with the so-called technology sector. Ongoing developments in electronics and computers tend to bring about changes at a rapid pace, and as such, the related businesses and commercialized products transform themselves both quickly and constantly. The chemical engineering contributions — such as scaleup, plant design, plant operation, and distribution — that are required to move those promising new developments toward commercial-scale production are often invisible. For example, for smartphones, the development and mass production of better materials for the screen and body, improved electronic materials, and longer-lasting batteries require many chemical engineering innovations.

Furthermore, the general public does not understand the key role chemical engineering innovation plays in providing food, water, energy, and goods to consumers. To remain viable and competitive, the CPI must innovate

every day — beyond just standard research and development (R&D) activities.

Companies throughout the CPI cannot afford to ignore opportunities for innovation. This article explains why organizations must make innovation a priority and discusses how companies can rethink their innovation strategies — moving beyond the pursuit of new products and services. A more-holistic view of innovation considers the entire consumer experience and the company's image and workforce.

The true meaning of innovation

A common myth is that innovation requires a lot of money and therefore is too expensive to be a priority. Yes, innovation in its most classical sense — *e.g.*, the development of a new advanced material, a new agrochemical product, or an innovative pharmaceutical therapy — is both expensive and time-consuming. Beyond the significant upfront R&D work, such efforts involve extensive testing, process and process control development, regulatory approvals and permitting, scaleup, distribution channel identification, etc. All these steps require significant capital and involve risks. As such, innovative new products are often pursued primarily by larger CPI companies that have the required resources, and efforts are often focused on those products that have enough potential to justify the significant investment and potential risk.

Innovation in the most classical sense is often limited to

those companies that have sufficient resources. However, a broader, more holistic definition of innovation — one that covers the entire customer experience — offers many opportunities for improvement that do not require deep pockets. To remain competitive, companies must innovate in every area, such as improving work processes, addressing challenges in manufacturing, attracting and retaining top talent, interacting with customers, encouraging the best contribution from every employee, and continuously improving overall operations (1). Such efforts are less capital-intensive and more abstract, but offer many opportunities to contribute favorably to the company's bottom line.

Innovation is not just about good ideas — it is about ideas that are practical and commercially viable, enabling broad implementation. For example, if a chemical reaction needs two costly reactants to produce an inexpensive product, then such a chemical reaction would be possible from a scientific perspective, but not practical from a business perspective.

Likewise, there is a key difference between invention and innovation. An invention is a new product, a new process, or a new technology, while innovation is something that serves a customer need in a new way (which, in some cases, may be an invention) (2). Therefore, innovation does not require an invention.

Innovation and the customer

In general, innovation involves generating value for the customer by understanding and satisfying their needs. In certain cases, new products are required to satisfy customer needs, but often, they can be satisfied simply by modifying existing products, reducing the price of the product, improving customer service, or improving purchasing, delivery, and servicing options. These innovation opportunities can provide greater value to the customer and foster customer satisfaction and corporate loyalty. The need to limit the cost of a product often drives efforts to reduce production costs (a key objective in many CPI operations) so that the savings can be passed on to customers.

Companies that do not have close contact with their customers may find it difficult to innovate. Many companies claim they are customer-focused, bombarding customers with surveys asking about the customer experience after they buy something or interact with the services they provide. However, such surveys (*i.e.*, voice-of-the-customer techniques), may not yield the desired result of better understanding of the customer's experience. Although voice-of-the-customer methods can be useful, many companies rely too much on them. These methods lend themselves more to incremental improvements than to breakthroughs (2). To come up with innovative solutions, companies must know not only what customers want, but why. Understanding the motivations behind customer behavior can inform innovative solutions.

The business case for innovation

In any industry, if a company is not able to innovate fast enough to better satisfy customers and remain ahead of the competition, then the company risks becoming obsolete. Some companies that were once the cornerstones of their industries have disappeared, in some cases because they focused too much on short-term objectives and neglected what made them successful in the first place — innovation.

Some analysts think that significant consolidation in the chemical, petrochemical, and oil and gas industries over the past 15–20 years is partially due to the inability of some players to evolve and offer timely innovations. A decade ago, for example, the oil and gas industry seemed (at least to outsiders) to have very few avenues for innovation. Then, the fracking revolution created opportunities to access reservoirs that had previously been inaccessible.

“For almost 90% of CEOs, generating organic growth through innovation has become essential for success in their industry,” writes Nicholas Webb in *The Innovation Playbook* (2). Hargadon and Sutton (3) agree: “Without a constant flow of ideas, a business is condemned to obsolescence.” As chemical engineers, we need to consider whether our discipline is moving fast enough to keep up or whether we have stagnated.

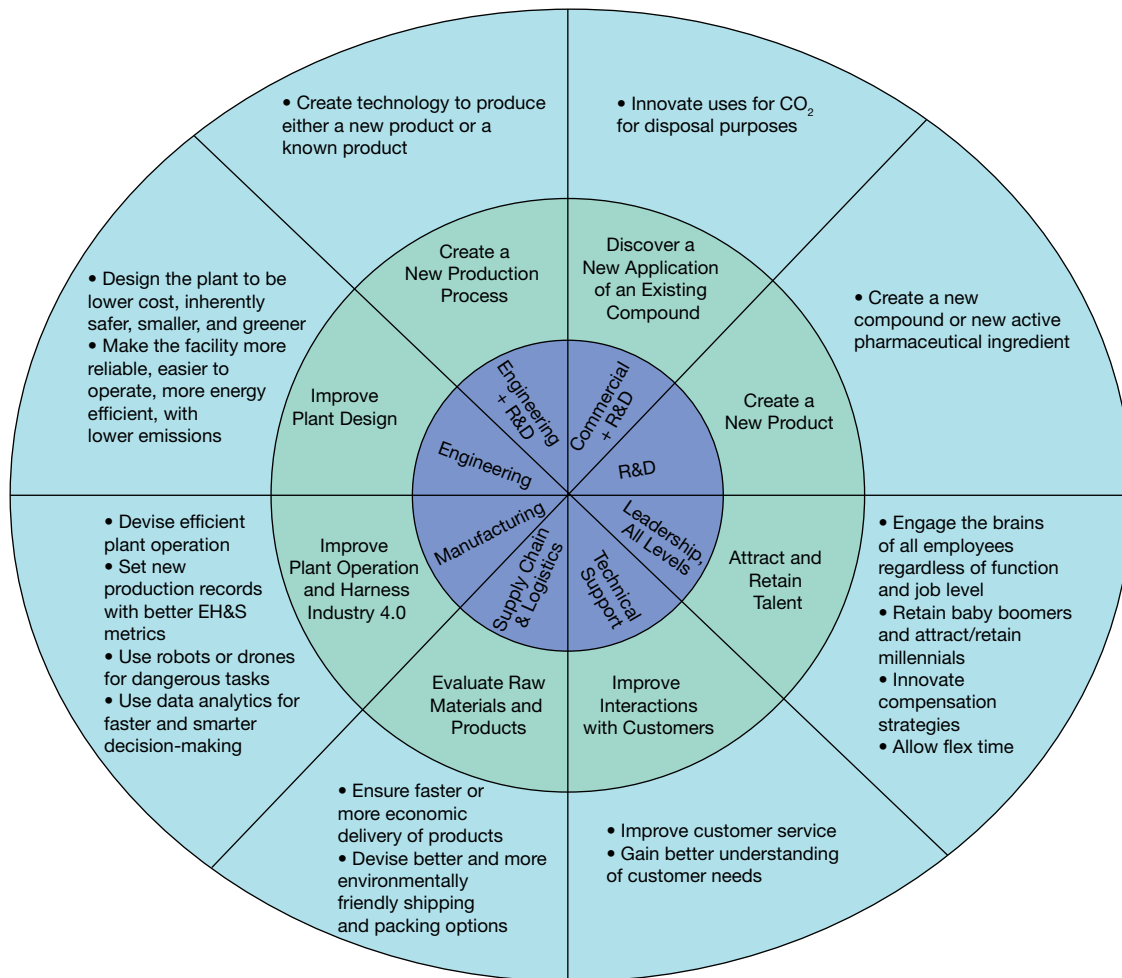
Innovation within chemical engineering

Chemical engineers are in an ideal position to develop solutions for some of today's most important problems, such as providing food, potable water, goods, and energy to a growing population (4). Satisfying these needs via conventional methods would be impossible and potentially disastrous for the planet. Significant advances are needed to improve productivity and yield of crops, reduce environmental contamination, improve energy and water stewardship, and more.

The CPI are generally not considered as innovative as the electronics, communications, and technology sectors. Investor Peter Thiel argues that “the amazing advances we have seen in computer science and communications have masked ominously disappointing progress in energy, transportation, biotech, disease prevention, and space travel” (5). Fortunately, via Industry 4.0, the CPI are now innovating in areas such as the industrial internet of things (IIoT), analytics using big data (which enable faster and smarter decision-making), broader connectivity via the cloud, advanced robotics, and improved machine-to-machine communications.

Some specific examples of chemical engineering solutions that have great potential for wider application, and thus are avenues for potential innovation, include:

- process intensification, *i.e.*, any chemical engineering development that leads to a substantially smaller, cleaner, and more energy-efficient technology (6)



▲ **Figure 1.** Opportunities abound for innovation in the chemical process industries (CPI). In the innermost (purple) circle are a variety of job functions a chemical engineer may fulfill at their company. The next (green) circle shows innovation opportunities. The outermost circle (blue) gives more specific examples of how to go about innovating.

- alternative/clean energy systems
- alternative production processes and chemical reactions that can reduce waste generation, consume less water, require less dangerous materials, and result in greater energy efficiency or higher yields of the desired product
- automation systems that can optimize production and quality and reduce production costs
- nanotechnology and nanomaterials
- advanced pharmaceuticals
- more-sustainable building and construction materials
- drones and robots to minimize or avoid risks to people associated with handling hazardous chemicals and work in confined spaces or elevated locations (*e.g.*, for flare inspections).

Chemical engineers throughout the CPI can innovate in their day-to-day work — regardless of the type of role or job they have, whether in R&D, manufacturing, engineering, supply chain, marketing, etc. (Figure 1).

Energy sector: We hear about new innovative solutions in the energy sector almost daily. Unfortunately, some of these are just ideas and inventions, not innovations. They may have validity from the strictly scientific viewpoint, but do not make much sense from a practical viewpoint. For example, the fact that methane and other chemicals and fuels can be produced using CO₂ as raw material does not mean that doing so in a lab or in a pilot plant will become the solution for CO₂ disposal or a great opportunity for the energy sector, as some of such ideas are advertised. For an invention to become a valid solution, it needs to meet some stringent criteria regarding economics, practicality, thermodynamics, etc. Likewise, many of these inventions do not make sense from an economic viewpoint because they require a tremendous amount of resources for industrial-scale implementation.

Unfortunately, the general public often does not understand the science and engineering concepts behind energy

production. Chemical engineers have the opportunity to help educate the general public about what is practical and feasible. They can also make contributions to the energy sector through their work in developing:

- alternative fuels
- batteries that are lighter, have more storage capacity, etc.
- solar panels that are less expensive, more efficient, more economical, easier to install, etc.
- lighting elements that are energy efficient and durable
- materials for transportation applications that are stronger and lighter
- insulation for housing and commercial buildings that is more efficient.

Food and agriculture. The chemical industry has played a major role in the food production and food conservation sector for more than a century. For example, without the use of ammonia-based fertilizers, humankind would only be able to provide food for a fraction of the world's current population.

Opportunities in agricultural sciences go much further than fertilizers and pesticides. Innovation will be driven by the increasing demand for food by a fast-growing population coupled with an increasing demand for beef and other animal proteins in the developing world. Food shortages are a real and pressing problem in many parts of the globe. In fact, according to the Food and Agricultural Organization (FAO), more than 840 million people go without enough to eat every day.

Food waste is another pressing concern. Every year, the U.S. throws out \$180 billion worth of food, which ends up in landfills and creates the greenhouse gas methane, which can cause environmental harm (7).

Feeding a growing population that is expected to surpass 9 billion by 2050, while minimizing negative environmental impacts, will require major innovations from chemical engineers. These goals will only be achieved via technologies that significantly improve land productivity and minimize food waste.

Barriers, challenges, mistakes, and misconceptions

A major barrier to innovation is the tendency of some companies and corporate leaders to play to not lose instead of playing to win. Corporate culture tends to be risk-adverse and hierarchical, which tends to stifle innovation. This environment can make personnel feel intimidated or reticent to call attention to problems and opportunities for improvement.

Creativity involves provocation, exploration, and risk-taking (8). The creative process typically involves some amount of trial and error, so a certain degree of failure must not only be tolerated, but encouraged. The old adage “you learn more from your failures than from your successes” is relevant in the quest for innovation. In certain organiza-

tions, “people who fail in experiments are often viewed as incompetent and that attitude can lead to counterproductive behavior,” writes Stefan Thomke in a *Harvard Business Review* article “Enlightened Experimentation: The New Imperative for Innovation” (9). Failures, however, should not be confused with mistakes. Mistakes produce little new or useful information and are therefore without value (9).

A big challenge for corporations is to achieve good results from quarter to quarter, while simultaneously investing in a strong innovation pipeline to help achieve results in the long term. Because globalization has generated fierce

OPPORTUNITIES TO LEARN FROM CHALLENGES AND FAILURES

Learn from failure, quickly. Thomas Edison famously said “I have not failed. I've just found 10,000 ways that won't work.” We learn everyday what does not work on countless projects and in many situations. The secret is to find out quickly what does not work — that is, fail quickly — instead of spending a lot of money and time before discovering that an idea does not work, and then learn from that failure.

Seek opportunities in manufacturing. Those who are not experienced in manufacturing roles often assume that manufacturing offers very little room for innovation. In reality, the day-to-day challenges in the manufacturing arena require ingenuity and innovation to keep plants running safely and continuously generating the desired output or yield. Such day-to-day challenges range from equipment failures, to raw material issues, to utility problems, and more. Manufacturing personnel must keep safety on their minds at all times. Innovation does not mean taking shortcuts and putting people and the environment at risk. Many chemical engineers in the manufacturing sector are unrecognized heroes and day-to-day champions of innovation.

Get more from capital projects and value engineering. Engineers who design chemical process plants and other capital-intensive projects may initially select state-of-the-art materials of construction, automation systems, complex equipment, and other aspects of the plant that produce a so-called gold-plated design. Then, a cost estimate is prepared and reality strikes — the beautiful design may not be economically feasible. If the project is not halted at that point, the typical next step is to perform a value engineering exercise that, if conducted well, can generate excellent and innovative ideas to reduce cost. Of course, the best approach would be to encourage more innovation at the beginning of any capital project, including early and proactive value-engineering exercises. However, this approach is not always adopted, and occasionally a capital project will require dramatic cost reductions. Some of these projects, in which the teams were forced to innovate due to smaller-than-expected budgets, become very successful.

competition in all industries, companies sometimes focus on the short-term financial results to satisfy some shareholders. However, in competitive environments, the only way to break away from the pack is through innovation. From the investors' viewpoint, companies with a strong culture and pipeline of innovation offer a better investment prospect.

Cultural gaps. Innovation can often be facilitated through collaboration with universities, entrepreneurs, startup companies, professional career coaches, and facilitators. The objective is to maximize diversity of thinking rather than limiting the conversation to people from the same company. This sort of open collaboration is relatively common and very productive in the pharmaceutical industry (10). However, in addition to valid intellectual property concerns, a common obstacle for open collaboration is cultural: members of one organization may reject ideas from another organization simply because of where those ideas originated. Such an attitude tends to be inefficient and ultimately hampers innovation (3).

Excessive use of tools can stifle innovation. The creative process often involves capturing ideas from multiple sources and developing those ideas. Developing an idea includes discussion to improve and expand the idea, consideration of alternative applications, and development of alternative and complementary ideas, as well as iterating these steps until the idea is fully developed. Many techniques and tools are available to generate and improve ideas, including classics like brainstorming sessions and

the Six Thinking Hats (8) methodology.

However, companies need to avoid falling into the trap of relying excessively on tools or methods. For instance, mandating personnel to follow many (sometimes boring) steps of a given method can stifle creativity.

Reference 2 reports that there are more than 170 different innovation management systems, but nearly every method is “a cookie-cutter approach that almost always results in failure.” Almost by definition, the creativity process requires thinking out of the box. Specifically, methods that require completing many stage gates, filling out multiple forms or spreadsheets, or gathering excessive amounts of data can overcomplicate and stifle innovation efforts (2).

Moving toward innovation

From an individual perspective, the best way to be innovative might be to keep your eyes and mind open. Of course, that is often easier said than done. The biggest enemy of innovation is conformism — the tendency to accept and maintain the status quo without trying to make things better.

Innovation requires a willingness to change and the assertiveness to decide what to change and what not to change. A classic example is summarized by Henry Ford, who said, “If I had asked the market what they wanted, they would have said a faster horse” (2).

Corporate innovation. While innovation has become a buzzword, the big question remains: How to get companies in the innovation mindset. Some companies have created high-ranking positions, appointing a Chief Innovation Officer to boost their innovation capabilities (1). Many companies have also engaged innovation gurus and implemented a variety of innovation programs. Other avenues to innovation include:

- promoting diversity of thought
- allowing some experimentation
- becoming more tolerant of failure
- adding innovation to the performance appraisal criteria
- creating a culture that not only promotes generation of ideas, but that quickly acts on those ideas (1).

These are all promising initiatives. The challenge is to implement them into the corporate culture and day-to-day work.

To drive innovation, companies must engage everyone. That is easy to say but not easy to achieve. Everyone does not include only engineers and leaders, but operators, maintenance personnel, plant management, etc.

Too often, workers who complete valuable but repetitive work are not encouraged to engage in innovative thinking. If employees are not engaged, they generally do not feel authorized to make suggestions that may help to address lingering problems or issues in the workplace. If attitudes like “it’s not my problem” or “it has been like

ENGAGE ALL EMPLOYEES IN INNOVATION EFFORTS

There are many reasons for involving employees at all levels — from the top floor to the shop floor — in innovation efforts:

- In today’s competitive environment, you need everybody’s best effort and perspective to succeed. Your competitors may already have mechanisms in place to fully engage their employees.
- The person doing the job is typically the most qualified to find ways to be more productive and to think of ways to yield a better product or provide better customer experience.
- There are countless opportunities to innovate (or at least improve) in all jobs, including some of the less glamorous ones.
- Because most people are creative under the right circumstances, limiting innovation to just engineers and scientists represents a tremendous waste of talent that companies cannot afford to squander.
- CPI companies face fierce competition for talent, but can attract and retain the best by offering the most interesting and satisfying jobs, which should include opportunities and encouragement to innovate.

that for 20 years” prevail, then there is little opportunity for innovation. Employees with the right motivation and attitude are more likely to create value for their companies.

Closing thoughts

Chemical engineers are in the best position to solve, or help to solve, some of the biggest problems of the century. Opportunities for innovation are present in all chemical engineering roles, not limited solely to positions in R&D or technology development. Chemical engineers who are willing to accept the innovation challenge are likely to have more interesting and rewarding careers than their counterparts, and they are more likely to better contribute to their companies and society.

So, what are you going to do today, this week, this quarter, and this year to be more innovative?

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